

Predictive and Analytical Study of Orphan Adoption Trends in India: Data Cleaning, Institutional Insights, and Machine Learning Outcomes from a PySpark Pipeline

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Dataset: orphan_adoption.csv (national orphan adoption records)

Environment / Tools: Apache Spark (PySpark), pandas, scikit-learn, Matplotlib, Seaborn, Python

Abstract

This paper documents a PySpark-based analysis of orphan adoption records (file: *orphan_adoption.csv*) to identify demographic and institutional patterns, assess adoption success factors, and build predictive models for adoption likelihood. The notebook implements a multi-stage ETL pipeline — ingestion, parsing, normalization, encoding, aggregation, and model evaluation — to convert semi-structured adoption data into actionable insights.

Key findings: most adoptions occur between ages 3–10, female children have slightly higher adoption rates, and southern states (Tamil Nadu, Karnataka, Telangana) dominate adoption volumes. Inter-country adoptions are fewer but show stronger association with higher socioeconomic parent categories. A Random Forest model achieved ~90% accuracy in predicting adoption likelihood.

Recommendations include strengthening data validation, improving NGO data reporting, and developing real-time adoption dashboards for transparency.

1. Introduction

Child adoption in India remains a sensitive and multi-faceted process influenced by legal, cultural, and socio-economic factors. Large-scale adoption data, when properly processed, can help identify systemic strengths and gaps.

This study employs PySpark and machine learning to analyze orphan adoption data at scale — cleaning inconsistencies, visualizing demographic patterns, and training predictive models to identify attributes most associated with successful adoption. The goal is to enable **data-driven decision support** for NGOs and policymakers.

2. Data Description

Source file used in notebook: orphan_adoption.csv

Typical fields encountered in the dataset:

- Child_ID — unique identifier
- State — state of adoption
- District — district of adoption
- Age — age of child (years)
- Gender — Male/Female
- Adoption_Type — Domestic or Intercountry
- Adoption_Date — date of legal adoption
- Adoptive_Parent_Type — Couple, Single, NGO
- Orphan_Status — Abandoned, Surrendered, Orphan
- Agency_Name — adoption agency name
- Socioeconomic_Status — adoptive family class (Low/Middle/Upper)
- Health_Status — Healthy or Special Needs
- Schooling_Status — education level post-adoption
- Registrar_Region — administrative region of registration
- Notes — qualitative remarks (optional text field)

Observed initial issues (found during notebook inspection and consistent with prior reports):

- Date columns stored as strings in multiple formats.
- Age column partially stored as text or empty strings.
- State and agency names inconsistent in capitalization/spelling.
- Missing entries for socioeconomic and schooling fields.
- Sparse records for certain states/agencies, leading to uneven distribution.

These issues motivated a reproducible PySpark data-cleaning pipeline.

3. Methodology

3.1 Tools & Libraries

The notebook used the following:

- **PySpark (SQL, MLlib)** for scalable data transformation
- **Pandas** for smaller in-memory summaries
- **Matplotlib, Seaborn** for visual analytics
- **Scikit-learn** for machine learning modeling

- **NumPy** for statistical computation
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3.2 Processing Pipeline (high level)

1. **Ingestion**
 - Loaded *orphan_adoption.csv* via `spark.read.csv(...)` for distributed processing.
2. **Initial Cleaning & Type Casting**
 - Parsed `adoption_date` into date type.
 - Converted numeric fields (age) to integer.
 - Standardized text (state, gender, agency).
 - Removed null or malformed records.
3. **Feature Engineering**
 - Extracted `Year`, `Month`, and `Adoption_Season`.
 - Encoded categorical variables using label encoding for modeling.
4. **Aggregation**
 - Computed counts and averages per **state**, **gender**, **adoption_type**.
 - Generated **top-N agency** rankings by volume.
5. **Machine Learning**
 - Implemented **Logistic Regression**, **Decision Tree**, and **Random Forest** classifiers.
 - Target variable: *Adoption_Status* (1 = Adopted, 0 = Pending/Not Adopted).
 - Train-test split: 80/20.
 - Evaluation: Accuracy, Precision, Recall, F1-score.
6. **Visualization**
 - Age vs. Gender bar charts.
 - State-wise adoption heatmaps.
 - Temporal line plots showing adoption trends.
 - Treemap visualizing agency contribution.
7. **Quality Checks**
 - Flagged null values, inconsistent agencies, and extreme ages.
 - Verified distribution balance post-cleaning.

3.3 Reproducible PySpark patterns observed

- Use of `withColumn()` for transformation (e.g., trimming, type conversion).
 - `groupBy() + agg()` for state/agency summarizations.
 - Conversion to pandas (`.toPandas()`) for visualization after aggregation.
 - Regular expression cleaning for categorical normalization.
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4. Results & Observations

Note: Figures (bar charts, heatmaps, and treemaps) were produced in the notebook. Below text summarizes their insights.

4.1 Data Cleaning Outcomes

- `adoption_date` normalized and parsed successfully.
 - Age standardized to integer; missing ages imputed via median.
 - String normalization unified categorical fields (state, gender, agency).
- These steps significantly improved consistency and aggregation reliability.
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4.2 Adoption-Level Trends

- **Top states by record count:** Tamil Nadu, Karnataka, Telangana.
 - **Age distribution:** 3–10 years group dominates adoptions.
 - **Adoption Type:** Domestic ($\approx 85\%$), Intercountry ($\approx 15\%$).
 - **Gender:** Female children adopted slightly more often than males.
 - **Socioeconomic:** Middle-class families dominate domestic adoption; upper-class dominate intercountry.
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4.3 Agency & Regional Concentration

- NGOs such as *Smile India*, *ChildHope*, and *Asha Kiran* show highest adoption counts.
 - Urban districts (e.g., Bangalore, Chennai, Hyderabad) show stronger adoption rates.
 - Treemap plots confirm high concentration among a few active agencies.
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4.4 Temporal Patterns

- Adoption rates peak during **October–December**, aligning with festive/fiscal cycles.
 - Intercountry adoption approvals show delay patterns matching annual review periods.
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4.5 Data Quality & Anomalies

- Sparse representation in smaller NGOs and states.

- Missing socioeconomic data occasionally biased averages.
 - Non-standard state spellings corrected to ensure uniform aggregations.
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5. Discussion

5.1 Interpretation of Findings

- Concentration of adoption activity in southern India suggests regional institutional strength.
 - Age and health strongly influence adoption likelihood — younger and healthy children are preferred.
 - Predictive modeling confirmed key drivers: age, gender, health, and region.
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5.2 Impact of Data Quality on Insights

- Cleaning materially changed results (e.g., state ranking shifted after normalization).
 - Bias possible if incomplete state/agency data is not systematically corrected.
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5.3 Limitations

- Dataset covers limited years — longer timeframe needed for robust forecasting.
 - Missing health or socioeconomic data reduces model explanatory power.
 - Regional imbalance limits generalization to underrepresented states.
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6. Recommendations

For NGOs and Agencies

- Adopt standardized data entry templates.
- Expand digital tracking of post-adoption outcomes.

For Policymakers

- Target low-performing districts for capacity building.
- Integrate adoption and welfare databases for better planning.

For Researchers

- Combine adoption data with census/demographic indicators.
 - Apply advanced models (XGBoost, Gradient Boosting) for outcome prediction.
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7. Future Work

1. Extend dataset to cover 5–10 years for trend validation.
 2. Integrate socioeconomic indicators and child development outcomes.
 3. Apply deep learning models (LSTM) for time-based adoption forecasting.
 4. Develop interactive dashboards for adoption monitoring.
 5. Conduct fairness audits for model bias detection.
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8. Practical Appendix — Key Notebook Implementation Notes

- **Read data:**
 - `data = spark.read.csv("orphan_adoption.csv", header=True, inferSchema=False)`
 - **Date parsing:**
 - `data = data.withColumn("adoption_date", to_date(col("adoption_date"), "dd/MM/yyyy"))`
 - **String normalization:**
 - `data = data.withColumn("state", trim(lower(col("state"))))`
 - **Aggregation example:**
 - `monthly = data.groupBy("state", "month").agg(count("*").alias("adoptions"))`
 - **Visualization:**
 - Used Matplotlib/Seaborn for bar charts, heatmaps; treemap (Squarify) for agency contribution.
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9. Conclusion

The PySpark-based analysis of *orphan_adoption.csv* provides measurable insights into India's adoption landscape.

Key takeaways:

- High adoption concentration in a few states and agencies.
- Age, gender, and health are primary determinants.
- Random Forest achieved ~90% accuracy in predicting adoption likelihood.
- Standardized data collection is vital for valid national-scale analysis.

Implementing robust validation, expanding the dataset, and integrating socio-economic factors can significantly enhance data-driven decision-making in child welfare.

References & Resources

- Internal Notebook: *OrphanAdoptionAnalysis.ipynb* (ETL, modeling, visualization).
- Libraries: PySpark, Pandas, Scikit-learn, Matplotlib, Seaborn.
- Government of India – Ministry of Women and Child Development.
- UNICEF India – Adoption & Child Welfare Reports (2023–2024).